

DEBMALYA PAL

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PROFESSIONAL SUMMARY

Software engineer with over 3 years of industry experience in building and maintaining scalable enterprise-grade applications, currently pursuing MS in Computer Science at UC San Diego.

EDUCATION

* University of California, San Diego (US) Master of Science – Computer Science	Sept. 2025 – Mar. 2027 GPA: 4.0
* University of Engineering and Management, Kolkata (IN) Bachelor of Technology – Computer Science	Jul. 2018 – Jun. 2022 GPA: 4.0

WORK EXPERIENCE

- * **Software Engineer – Cognizant (Client: NextEra Energy)**
Pune, Maharashtra, India Apr. 2025 – Sept. 2025
- Modernized a stand-alone asset management system to RESTful web app for world's largest producer of wind and solar energy.
 - Automated critical data synchronization via cron jobs, ensuring high data integrity and reducing manual efforts by 75%.
- Tech Stack:** Java, Spring Boot, Spring JDBC, React, Oracle, REST APIs, Docker, CI/CD
- * **Software Engineer – Cognizant (Client: Major US InsurTech Marketplace)**
Pune, Maharashtra, India Sept. 2022 – Mar. 2025
- Engineered core modules for a B2B2C digital insurance platform supporting 10,000+ daily active users.
 - Migrated REST endpoints to GraphQL, reducing network payload size by 50% and significantly improving frontend latency.
 - Architected ETL pipelines on AWS processing 1.5M+ records monthly, eliminating 95% of manual intervention.
 - Implemented GenAI (LLM) solutions to automate disaster recovery documentation, slashing manual preparation time by 35%.
- Tech Stack:** Java, Spring Boot, Scala, Play, React, TypeScript, PostgreSQL, Python, AWS (S3, RDS, ECR, EKS, Lambda, SQS), REST APIs, GraphQL, Docker, Azure OpenAI

TECHNICAL SKILLS

- * **Languages:** Java, Python, JavaScript, TypeScript, SQL, HTML, CSS
- * **Frameworks & Libraries:** Spring Boot, Spring, React, Flask, Streamlit
- * **Architecture & APIs:** Microservices, REST API, GraphQL, JSON, XML, Swagger
- * **Databases & Storage:** PostgreSQL, MySQL, Oracle, MongoDB (NoSQL), AWS RDS, DynamoDB, Redis (ElastiCache)
- * **DevOps & Cloud:** Git, Docker, Kubernetes, AWS (S3, Lambda, RDS, ECR, EKS, SQS), CI/CD
- * **Testing:** JUnit, Mockito, Jest, Postman
- * **Machine Learning:** Scikit-Learn, Pandas, NumPy, Matplotlib, NLTK, Regression, Classification, Natural Language Processing
- * **Generative AI:** Large Language Models (LLMs), RAG, Embeddings, LangChain, Vector DBs (Qdrant), Azure OpenAI, MCP
- * **Collaboration & Workflows:** Agile, Scrum, Jira, Confluence, Parabol

PERSONAL PROJECTS

- * **MediPredict: ML Powered Disease Prediction System**
- Built an ML system that predicts diseases using classification algorithms based on user demographics and symptoms.
 - Designed an interactive web interface for symptom input and prediction results.
- Tech Stack:** Python (NumPy, Pandas, Scikit-Learn), Flask, React
- * **CovVaxPulse: Sentiment Analysis - Covid19 Vaccination Drive**
- Built a data ingestion pipeline to collect vaccination-related data from social media, with scheduled jobs for periodic data retrieval.
 - Conducted sentiment analysis on vaccination-related tweets to assess public perception and its evolution over time.
 - Built interactive Streamlit dashboards to visualize sentiment trends and enable exploratory analysis of opinion shifts.
- Tech Stack:** Python (NumPy, Pandas, NLTK, TextBlob, Scikit-Learn, Matplotlib), Streamlit, PostgreSQL, AWS (EC2, RDS)
- * **Health & Fitness Tracker Application**
- Developed a full-stack fitness app for tracking vitals such as BMI, sleep, hydration, and workouts.
 - Added goal-setting features and personalized progress dashboards.
- Tech Stack:** Python, Flask, React JS, MySQL
- * **RxFlow - Microservices Order Management System**
- Built a web-based platform for scheduling, reordering, and purchasing prescription medications.
 - Implemented repeat prescriptions, order tracking, and automated delivery scheduling.
- Tech Stack:** Java, Spring Boot, Spring Data JPA, Spring Security, React, Microservices, MySQL, Docker, AWS (RDS, EBS)